

Supplementary information for “Graph-conditioned state-space models for genetic perturbation response prediction: a controlled evaluation against simple baselines”

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S1 Legacy perturbation encoding

Table S1 quantifies the encoding flaw described in Section 3.1 of the main text, computed with the original data loader (legacy/data_loader.py) and split manifests (splits/splits_manifest_*.json) by gcpm/audit_hvg_encoding.py. HVGs were selected on training cells only, as in the original pipeline.

Table S1: Legacy HVG-restricted encoding (Section 3.1): number of the 105 CRISPRa targets among the 500 HVGs, and perturbations whose input vector is all zero, for the ten split manifests of the earlier version of this work.

Split	Targets in HVGs	Train all-zero	Distinct train inputs	Test all-zero	Test n
0	17	133/169 (79%)	16	52/66 (79%)	66
1	12	146/183 (80%)	14	49/52 (94%)	52
2	14	147/174 (84%)	14	43/61 (70%)	61
3	13	133/161 (83%)	14	60/74 (81%)	74
4	13	134/167 (80%)	15	61/68 (90%)	68
5	14	132/168 (79%)	17	59/67 (88%)	67
6	14	133/161 (83%)	15	58/74 (78%)	74
7	15	136/177 (77%)	19	50/58 (86%)	58
8	14	137/167 (82%)	14	58/68 (85%)	68
9	14	135/176 (77%)	19	55/59 (93%)	59

S2 Model development on the validation set of split 1

All design decisions were made on the validation perturbations of split 1; test perturbations were not used. Table S2 lists the successive changes and the lowest validation MSE (all genes) reached in short pilot runs (12–35 epochs). For reference, predicting no change gives 0.00433, the train mean 0.00361, the linear model 0.00312 and the additive baseline 0.00244 on the same perturbations (the latter using the training singles only).

S3 Statistical comparisons and initialisation variance

Table S3 lists all pairwise tests between GCP-Mamba and the other models on the Norman data (deep models averaged over initialisations). Table S4 shows how much the split-level error of each model changes between initialisations; the within-split range is of the same size as the differences between graph-conditioned and graph-free variants. Table S5 gives the Pearson correlation on the top-20 DE genes for every model.

Table S2: Pilot runs of GCP-Mamba on the validation perturbations of split 1.

Change	Best validation MSE	Observation
Initial model (scalar readout, raw PCA loadings)	0.00362	equal to the train mean; training loss also at train-mean level
+ bilinear readout	0.00360	unchanged
+ standardised gene embeddings	0.00569	fits training data, generalises worse than the mean
+ per-target embeddings, dropout, weight decay 10^{-2}	0.00523	large initial loss (random outputs on all genes)
+ initialisation at the train-mean prediction ($\text{lr} = 2 \times 10^{-3}$)	0.00320	first improvement over the mean
+ learning rate 10^{-3} (final)	0.00297	

Table S3: GCP-Mamba versus every other model on the Norman data: two-sided Wilcoxon signed-rank tests on per-perturbation differences in MSE (top-20 DE genes), pooled over the five splits; Holm correction within each test group. Negative median differences favour GCP-Mamba.

Test group	Comparison model	n	Median diff.	P	P_{Holm}
All test	No change	561	-0.1704	1.1e-81	1.6e-80
All test	Train mean	561	-0.0523	7.6e-47	9.9e-46
All test	Additive	561	+0.0269	8.5e-23	9.4e-22
All test	Linear (PCA)	561	-0.0222	3.2e-12	2.5e-11
All test	Linear (ctrl emb.)	561	-0.0416	8.3e-46	9.9e-45
All test	GEARS	107	+0.0167	0.0013	0.0025
All test	GCP-Mamba (Δ only)	561	+0.0054	8.1e-07	4.8e-06
All test	GCP-Mamba (Δ only, $\mathbf{w}_c \sim \mathcal{N}(0, 1)$)	561	+0.0046	3.5e-05	0.00014
All test	GCP-Mamba (permuted graph)	561	+0.0013	5.3e-05	0.00016
All test	GCP-Mamba (random order)	561	+0.0064	1.5e-08	1e-07
All test	Mamba (no graph)	561	+0.0024	4.1e-06	2.1e-05
All test	Graph-MLP (no scan)	561	-0.0002	0.83	0.83
All test	GCP-Mamba (additive prior)	561	+0.0273	5.2e-22	5.2e-21
All test	Mamba (additive prior)	561	+0.0281	1.6e-21	1.4e-20
Unseen single	No change	185	-0.0540	1.7e-19	2.4e-18
Unseen single	Train mean	185	-0.0023	0.06	0.42
Unseen single	Additive	185	-0.0043	0.00087	0.0096
Unseen single	Linear (PCA)	185	-0.0010	0.46	1
Unseen single	Linear (ctrl emb.)	185	-0.0025	0.014	0.11
Unseen single	GEARS	36	+0.0518	0.002	0.02
Unseen single	GCP-Mamba (Δ only)	185	+0.0006	0.44	1
Unseen single	GCP-Mamba (Δ only, $\mathbf{w}_c \sim \mathcal{N}(0, 1)$)	185	-0.0003	0.35	1
Unseen single	GCP-Mamba (permuted graph)	185	-0.0010	0.0088	0.079
Unseen single	GCP-Mamba (random order)	185	+0.0014	0.15	0.74
Unseen single	Mamba (no graph)	185	-0.0000	0.28	1
Unseen single	Graph-MLP (no scan)	185	-0.0008	0.11	0.66
Unseen single	GCP-Mamba (additive prior)	185	-0.0049	2.6e-05	0.00034
Unseen single	Mamba (additive prior)	185	-0.0056	6.7e-05	0.0008
Double, 0/2 seen	No change	32	-0.2160	3.3e-08	4.6e-07
Double, 0/2 seen	Train mean	32	-0.0487	0.0034	0.044
Double, 0/2 seen	Additive	32	+0.0183	0.33	1
Double, 0/2 seen	Linear (PCA)	32	-0.0417	0.018	0.2
Double, 0/2 seen	Linear (ctrl emb.)	32	-0.0464	0.005	0.06

Table S3: GCP-Mamba versus every other model on the Norman data: two-sided Wilcoxon signed-rank tests on per-perturbation differences in MSE (top-20 DE genes), pooled over the five splits; Holm correction within each test group. Negative median differences favour GCP-Mamba.

Test group	Comparison model	n	Median diff.	P	P_{Holm}
Double, 0/2 seen	GEARS	9	+0.0941	0.055	0.55
Double, 0/2 seen	GCP-Mamba (Δ only)	32	+0.0084	0.3	1
Double, 0/2 seen	GCP-Mamba (Δ only, $\mathbf{w}_c \sim \mathcal{N}(0, 1)$)	32	+0.0038	0.41	1
Double, 0/2 seen	GCP-Mamba (permuted graph)	32	+0.0053	0.23	1
Double, 0/2 seen	GCP-Mamba (random order)	32	+0.0058	0.61	1
Double, 0/2 seen	Mamba (no graph)	32	-0.0019	0.75	1
Double, 0/2 seen	Graph-MLP (no scan)	32	+0.0006	0.83	1
Double, 0/2 seen	GCP-Mamba (additive prior)	32	+0.0172	0.32	1
Double, 0/2 seen	Mamba (additive prior)	32	+0.0067	0.59	1
Double, 1/2 seen	No change	254	-0.2765	1.8e-40	2.5e-39
Double, 1/2 seen	Train mean	254	-0.0995	3.9e-29	4.6e-28
Double, 1/2 seen	Additive	254	+0.0575	2.2e-21	2.2e-20
Double, 1/2 seen	Linear (PCA)	254	-0.0389	3.5e-08	2.8e-07
Double, 1/2 seen	Linear (ctrl emb.)	254	-0.0800	8.8e-30	1.2e-28
Double, 1/2 seen	GEARS	43	+0.0077	0.062	0.062
Double, 1/2 seen	GCP-Mamba (Δ only)	254	+0.0106	0.00029	0.0011
Double, 1/2 seen	GCP-Mamba (Δ only, $\mathbf{w}_c \sim \mathcal{N}(0, 1)$)	254	+0.0086	0.00071	0.0021
Double, 1/2 seen	GCP-Mamba (permuted graph)	254	+0.0046	2e-05	9.8e-05
Double, 1/2 seen	GCP-Mamba (random order)	254	+0.0127	4e-06	2.8e-05
Double, 1/2 seen	Mamba (no graph)	254	+0.0064	5.5e-06	3.3e-05
Double, 1/2 seen	Graph-MLP (no scan)	254	+0.0020	0.03	0.06
Double, 1/2 seen	GCP-Mamba (additive prior)	254	+0.0576	8.2e-22	9e-21
Double, 1/2 seen	Mamba (additive prior)	254	+0.0610	8.1e-21	7.3e-20
Double, 2/2 seen	No change	90	-0.3615	1.7e-16	2.4e-15
Double, 2/2 seen	Train mean	90	-0.2286	1.7e-16	2.4e-15
Double, 2/2 seen	Additive	90	+0.1065	6.4e-15	7e-14
Double, 2/2 seen	Linear (PCA)	90	-0.0720	2e-05	9.9e-05
Double, 2/2 seen	Linear (ctrl emb.)	90	-0.1615	4.2e-15	5.1e-14
Double, 2/2 seen	GEARS	19	+0.0015	0.57	0.57
Double, 2/2 seen	GCP-Mamba (Δ only)	90	+0.0157	9.1e-05	0.00036
Double, 2/2 seen	GCP-Mamba (Δ only, $\mathbf{w}_c \sim \mathcal{N}(0, 1)$)	90	+0.0239	7e-06	4.9e-05
Double, 2/2 seen	GCP-Mamba (permuted graph)	90	+0.0064	1.3e-05	7.5e-05
Double, 2/2 seen	GCP-Mamba (random order)	90	+0.0150	0.00035	0.0011
Double, 2/2 seen	Mamba (no graph)	90	+0.0175	5.6e-06	4.5e-05
Double, 2/2 seen	Graph-MLP (no scan)	90	-0.0032	0.1	0.2
Double, 2/2 seen	GCP-Mamba (additive prior)	90	+0.1045	7e-15	7e-14
Double, 2/2 seen	Mamba (additive prior)	90	+0.1063	7.7e-15	7e-14

S4 Hyperparameters and compute

Table S4: Variation due to random initialisation on the Norman data. For each split, the MSE (top-20 DE genes, all test perturbations) of three independently initialised runs; shown are the mean over splits of the within-split range and standard deviation, and the mean split-level MSE.

Model	Mean MSE	Within-split s.d.	Within-split range
GCP-Mamba	0.290	0.024	0.046
GCP-Mamba (permuted graph)	0.285	0.023	0.042
Mamba (no graph)	0.278	0.016	0.032
Graph-MLP (no scan)	0.288	0.017	0.033
GCP-Mamba (additive prior)	0.225	0.001	0.003

Table S5: Pearson correlation of predicted and observed expression change on the top-20 DE genes (mean $\pm$ s.d. over five splits; higher is better)

Model	Unseen single	Double, 0/2 seen	Double, 1/2 seen	Double, 2/2 seen	All test
No change	—	—	—	—	—
Train mean	0.53 ± 0.05	0.72 ± 0.12	0.66 ± 0.06	0.67 ± 0.07	0.62 ± 0.04
Additive	0.54 ± 0.05	0.70 ± 0.15	0.82 ± 0.03	0.97 ± 0.01	0.75 ± 0.03
Linear (PCA)	0.52 ± 0.07	0.64 ± 0.18	0.73 ± 0.03	0.84 ± 0.04	0.67 ± 0.03
Linear (ctrl emb.)	0.52 ± 0.05	0.70 ± 0.09	0.69 ± 0.06	0.76 ± 0.08	0.65 ± 0.05
GCP-Mamba	0.55 ± 0.08	0.71 ± 0.13	0.76 ± 0.03	0.86 ± 0.04	0.71 ± 0.04
GCP-Mamba (Δ only)	0.57 ± 0.06	0.71 ± 0.15	0.76 ± 0.02	0.87 ± 0.03	0.72 ± 0.03
GCP-Mamba (Δ only, $\mathbf{w}_c \sim \mathcal{N}(0, 1)$)	0.56 ± 0.07	0.72 ± 0.14	0.76 ± 0.03	0.88 ± 0.03	0.71 ± 0.04
GCP-Mamba (permuted graph)	0.53 ± 0.08	0.68 ± 0.16	0.76 ± 0.03	0.86 ± 0.03	0.70 ± 0.04
GCP-Mamba (random order)	0.58 ± 0.08	0.73 ± 0.12	0.77 ± 0.03	0.88 ± 0.02	0.72 ± 0.04
Mamba (no graph)	0.55 ± 0.08	0.70 ± 0.14	0.77 ± 0.03	0.87 ± 0.03	0.71 ± 0.04
Graph-MLP (no scan)	0.55 ± 0.06	0.68 ± 0.17	0.77 ± 0.02	0.86 ± 0.03	0.70 ± 0.03
GCP-Mamba (additive prior)	0.53 ± 0.06	0.70 ± 0.17	0.82 ± 0.03	0.98 ± 0.01	0.74 ± 0.03
Mamba (additive prior)	0.54 ± 0.06	0.70 ± 0.18	0.82 ± 0.03	0.98 ± 0.01	0.75 ± 0.04

Table S6: Hyperparameters shared by all deep-model variants.

Parameter	Value
Layers, model width d , state size N	2, 32, 4
Readout rank	16
Gene embedding	64 PCs of control-cell expression, standardised
Co-expression graph	$ r $ on control cells, $k = 20$ neighbours, symmetrised
Graph signal	3 diffusion steps, decay 0.5, $\log(1 + 10^3 \cdot)$
Scan chunk length, log-decay clamp	16, -5
Optimiser	AdamW, lr 10^{-3} , weight decay 10^{-2} , batch 8, clip 1.0
Dropout (perturbation encoder)	0.1
Epochs	min. 40, max. 100, patience 15 on validation MSE
Seeds	split s : s , $s + 1000$, $s + 2000$ (three initialisations for the main variants)
Hardware	4-core CPU, no GPU